

Notes: Garlappi and Song (JFE, 2017)

Capital utilization, market power, and the pricing of investment shocks

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1 Big picture

- **Question.** How do *capital utilization* and *market power* change the way investment-specific technology (IST) shocks are priced in financial markets?
- **Setting.** The paper builds a two-sector production-based asset-pricing model with a consumption sector, an investment sector, recursive preferences, endogenous capital utilization, and monopolistic competition among intermediate-good producers.
- **Why this matters.** Earlier IST papers often effectively assumed *fixed utilization* and *perfect competition*. Under those assumptions, positive IST shocks tend to lower the price of capital and therefore lower firms' values, which naturally pushes the market beta on IST shocks negative. Garlappi and Song ask whether that conclusion survives once firms can use existing capital more intensively and earn monopoly rents.
- **Main mechanism #1: utilization affects the *price of risk*.** With fixed utilization, a positive IST shock pulls labor into the investment sector, lowers current consumption, raises marginal utility, and therefore tends to make the market price of IST risk negative. With sufficiently flexible utilization, firms can raise current output by using installed capital more intensively, so the same shock can lower marginal utility and flip the price of IST risk positive.
- **Main mechanism #2: market power affects the *quantity of risk*.** Under perfect competition, firm value mainly equals the replacement cost of installed capital, so a positive IST shock that cheapens new capital lowers firm value and gives a negative IST beta. With market power, cheaper capital also raises the value of future monopoly rents, which can make the market beta on IST shocks positive.
- **Main mechanism #3: preferences matter for both.** The sign and size of IST pricing depend strongly on the elasticity of intertemporal substitution (EIS) and on whether households prefer early or late resolution of uncertainty. High EIS and preference for early resolution make positive IST prices of risk easier to obtain.
- **Main qualitative result.** The paper overturns the idea that IST shocks must be priced through a negative price of risk and a negative market beta. Both objects can be positive in an economically reasonable environment.
- **Main quantitative result.** In the benchmark monthly calibration with long-run risks in technology growth, market power, and high EIS, the model generates an annual log risk-free rate of about **0.50%** and an annual log equity premium of about **5.14%**, close to the data.
- **Risk-premium decomposition.** In the benchmark, only about **11%** of the market premium comes from short-run technology shocks; about **89%** comes from long-run components in TFP and IST growth. The single biggest component is long-run TFP risk.
- **Most important comparative-static result.** Market power and high EIS are quantitatively decisive. Without market power, the same calibration produces a *negative* market risk premium. Flexible utilization matters too, but its quantitative effect depends on whether one looks at period utility or continuation utility and on whether shocks are short-run or long-run.
- **Big takeaway.** To understand the asset-pricing implications of investment shocks, it is not enough to know that new capital becomes cheaper. One also has to know (i) whether firms can re-optimize the *use of old capital* and (ii) whether firms retain *rents* when technology improves.

2 Model environment and key objects

2.1 Households and recursive preferences

The representative household has Epstein–Zin preferences over consumption C_t and labor L_t :

$$U_t = \left[(1 - \beta)(C_t(1 - \psi L_t^\theta))^{1-\rho} + \beta \left(\mathbb{E}_t U_{t+1}^{1-\gamma} \right)^{\frac{1-\rho}{1-\gamma}} \right]^{\frac{1}{1-\rho}}. \quad (1)$$

- β is the discount factor.
- $1/\rho$ is the EIS.
- γ is relative risk aversion.
- ψ and θ control the level and curvature of labor disutility.

The household budget constraint is

$$P_t^C C_t = W_t L_t + D_t^C + D_t^I, \quad (2)$$

where W_t is the wage and D_t^C, D_t^I are dividends from the two sectors.

The one-period stochastic discount factor is

$$M_{t,t+1} = \beta \left(\frac{C_{t+1}}{C_t} \right)^{-\rho} \left(\frac{1 - \psi L_{t+1}^\theta}{1 - \psi L_t^\theta} \right)^{1-\rho} \left(\frac{V_{t+1}^{1-\gamma}}{\mathbb{E}_t[V_{t+1}^{1-\gamma}]} \right)^{\frac{\rho-\gamma}{1-\gamma}}, \quad (3)$$

where the last term is the continuation-utility component.

Interpretation.

- When $\rho = \gamma$, utility collapses to time-separable CRRA utility and the continuation-utility term disappears.
- When $\rho \neq \gamma$, shocks can be priced not just because they move *current* consumption, but also because they change *future utility*.
- This is crucial later: the effect of utilization flexibility on the price of risk is very different under recursive preferences with long-run risks than under simple CRRA preferences.

2.2 Final-good sectors and market power

There are two productive sectors:

- the **C-sector**, which produces the final consumption good;
- the **I-sector**, which produces the final capital good.

In each sector $J \in \{C, I\}$, a perfectly competitive final-good producer aggregates differentiated intermediate goods using CES technology:

$$Y_t^J = \left(\sum_{f=1}^{N^J} (x_{f,t}^J)^{\frac{\nu^J-1}{\nu^J}} \right)^{\frac{\nu^J}{\nu^J-1}}. \quad (4)$$

Demand for intermediate good f is

$$x_{f,t}^J = \left(\frac{p_{f,t}^J}{P_t^J} \right)^{-\nu^J} Y_t^J, \quad J \in \{C, I\}. \quad (6)$$

Interpretation.

- ν^J is the elasticity of substitution across intermediate goods.
- The gross markup is $\nu^J/(\nu^J - 1)$; equivalently, the net markup is $1/(\nu^J - 1)$.
- Perfect competition is the limit $\nu^J \rightarrow \infty$.
- Lower ν^J means more market power and bigger monopoly rents.

2.3 Intermediate-good production, utilization, and technology shocks

Intermediate firms produce with Cobb–Douglas technology:

$$y_{f,t}^C = A_t (u_{f,t}^C k_{f,t}^C)^{1-\alpha} (l_{f,t}^C)^\alpha, \quad (7)$$

$$y_{f,t}^I = A_t Z_t (u_{f,t}^I k_{f,t}^I)^{1-\alpha} (l_{f,t}^I)^\alpha. \quad (8)$$

- A_t is neutral total factor productivity (TFP).
- Z_t is investment-specific technology (IST); it improves the productivity of producing new capital goods.
- $u_{f,t}^J$ is the intensity with which installed capital is used.

Short-run growth in the shocks is

$$a_t = \bar{\mu}^a + \mu_{t-1}^a + \varepsilon_t^a, \quad (9)$$

$$z_t = \bar{\mu}^z + \mu_{t-1}^z + \varepsilon_t^z, \quad (10)$$

where $a_t = \log(A_t/A_{t-1})$ and $z_t = \log(Z_t/Z_{t-1})$.

In the quantitative section, the expected growth components follow long-run-risk processes:

$$\mu_t^a = \rho_{\mu^a} \mu_{t-1}^a + \varepsilon_t^{\mu^a}, \quad (29)$$

$$\mu_t^z = \rho_{\mu^z} \mu_{t-1}^z + \varepsilon_t^{\mu^z}. \quad (30)$$

Interpretation.

- The paper studies both short-run technology news and small persistent long-run components.
- Quantitatively, the long-run components do most of the work in generating a realistic premium.

2.4 Capital accumulation, depreciation, and adjustment costs

Capital evolves according to

$$k_{f,t+1}^J = k_{f,t}^J (1 + i_{f,t}^J - \delta(u_{f,t}^J)), \quad J \in \{C, I\}. \quad (11)$$

Depreciation rises with utilization:

$$\delta(u^J) = \delta_0 + \delta_1 \frac{(u^J)^{1+\xi^J} - 1}{1 + \xi^J}, \quad \xi^J > 0. \quad (12)$$

- A lower ξ^J means more flexible utilization.
- A higher ξ^J means using capital more intensively becomes sharply more expensive, so utilization is closer to fixed.

Investment is subject to Jermann-style convex adjustment costs, denoted $\varphi(i^J)$, with curvature parameter ϕ^J and normalized to be zero in steady state:

$$\varphi(i^J) \text{ is increasing and convex in } i^J, \quad \phi^J \geq 1. \quad (13)$$

Interpretation.

- Utilization is about how firms adjust the service flow from *old* capital.
- Investment is about how firms change the stock of *new* capital.
- The paper's key point is that these two margins have very different asset-pricing implications.

2.5 Firm problem, dividends, and market value

A representative intermediate-good firm in sector J solves

$$V_{f,t}^J = \max_{\{l_{f,s}^J, i_{f,s}^J, u_{f,s}^J\}_{s \geq t}} \mathbb{E}_t \sum_{s=t}^{\infty} M_{t,s} d_{f,s}^J, \quad (14)$$

subject to

$$d_{f,s}^J = p_{f,s}^J y_{f,s}^J - W_s l_{f,s}^J - P_s^I \varphi(i_{f,s}^J) k_{f,s}^J. \quad (1)$$

Aggregating within each sector,

$$V_t^J = \mathbb{E}_t \sum_{s=t}^{\infty} M_{t,s} D_s^J, \quad J \in \{C, I\}, \quad (15)$$

and the aggregate market value is

$$V_t^M = V_t^C + V_t^I. \quad (16)$$

A very important decomposition later in the paper writes firm value as

$$V_{f,t}^J = \underbrace{q_t k_{f,t+1}^J}_{\text{assets in place}} + \underbrace{\mathbb{E}_t \sum_{s=1}^{\infty} M_{t,t+s} \left[\frac{1}{\nu^J} p_{f,t+s}^J y_{f,t+s}^J \right]}_{\text{monopoly rents}}, \quad J \in \{C, I\}. \quad (32)$$

Interpretation.

- Under perfect competition, $\nu^J \rightarrow \infty$, monopoly rents vanish and value equals assets in place.
- With market power, firm value includes both installed capital and the discounted value of future markups.
- This is the cleanest way to see why market power can flip the sign of the market beta on IST shocks.

2.6 Market clearing and the aggregate state

The investment good is used to build capital in both sectors:

$$\sum_{f=1}^{N^C} \varphi(i_{f,t}^C) k_{f,t}^C + \sum_{f=1}^{N^I} \varphi(i_{f,t}^I) k_{f,t}^I = Y_t^I. \quad (17)$$

Because firms within each sector are symmetric in equilibrium, the model can be written in terms of sectoral aggregates K_t^J, L_t^J, Y_t^J . After rescaling, the economy is solved with four state variables:

- expected TFP growth, μ_t^a ;
- expected IST growth, μ_t^z ;
- relative capital stock, $k_t = \log(K_t^I / K_t^C)$;
- a rescaled productivity-capital ratio state for the I-sector.

2.7 Asset-pricing decomposition

Projecting the SDF on the technology shocks,

$$M_{t,t+1} = \mathbb{E}_t[M_{t,t+1}] - \frac{\lambda_t^a}{\sigma_a^2} \varepsilon_{t+1}^a - \frac{\lambda_t^z}{\sigma_z^2} \varepsilon_{t+1}^z, \quad (19)$$

so the price of risk for shock $x \in \{a, z\}$ is

$$\lambda_t^x = -\text{Cov}_t(M_{t,t+1}, \varepsilon_{t+1}^x). \quad (20)$$

For any asset j ,

$$R_{t+1}^j = \mathbb{E}_t[R_{t+1}^j] + \beta_{j,t}^a \varepsilon_{t+1}^a + \beta_{j,t}^z \varepsilon_{t+1}^z, \quad (21)$$

and therefore

$$RP_{j,t} = \beta_{j,t}^a \lambda_t^a + \beta_{j,t}^z \lambda_t^z. \quad (22)$$

For the market portfolio,

$$RP_{M,t} = \beta_{M,t}^a \lambda_t^a + \beta_{M,t}^z \lambda_t^z. \quad (23)$$

Interpretation.

- λ is the *price* of a shock: how the shock moves marginal utility.
- β is the *quantity* of a shock: how the shock moves market value.
- The whole paper can be summarized as follows:
 - utilization mainly changes λ^z ;
 - market power mainly changes β_M^z ;
 - preferences, especially EIS, matter for both.

3 Mechanisms and qualitative results (all equations + interpretation)

3.1 Fixed utilization: why the IST price of risk can be negative

Start with the simplest case: CRRA utility, fixed capital utilization, and perfect competition.

- A positive IST shock makes the investment sector more productive.
- Labor shifts from the consumption sector to the investment sector.
- Because installed capital cannot be used more intensively, current consumption falls.
- Marginal utility rises, so the SDF rises.

Therefore a positive IST shock is a *bad* state for the representative household, and

$$\lambda^z < 0.$$

Economic intuition.

- The household is willing to accept a lower expected return for assets that pay off when marginal utility is high.
- This is the classic logic behind the negative IST price of risk in models that resemble Papanikolaou's benchmark environment.

3.2 Flexible utilization: why the IST price of risk can turn positive

Now allow firms to choose u_t^J endogenously.

- A positive IST shock lowers the replacement cost of capital.
- Because replacing worn-out capital is now cheaper, firms are willing to run existing capital harder.
- Higher utilization raises current output in the consumption sector and offsets the labor reallocation effect.

If the utilization margin is sufficiently responsive, current consumption rises rather than falls. Then marginal utility falls and

$$\lambda^z > 0.$$

Interpretation.

- Utilization acts as an extra short-run adjustment margin.
- The sign of the IST price of risk is therefore *not* pinned down by the fact that IST shocks improve investment opportunities; it also depends on how easily firms can transform that shock into current consumption.

3.3 Perfect competition: why the IST market beta is negative

Under perfect competition, the monopoly-rent term in (32) is zero, so firm value equals assets in place:

$$V_{f,t}^J = q_t k_{f,t+1}^J.$$

A positive IST shock lowers the price of new capital q_t . Therefore the value of installed capital falls and market value declines:

$$\beta_M^z < 0.$$

Intuition.

- IST shocks are like favorable *supply shocks* for producing capital goods.
- That is good for future investment, but bad for the value of old capital already sitting on firms' balance sheets.

3.4 Market power: why the IST market beta can turn positive

With monopolistic competition, firm value contains monopoly rents:

$$V_{f,t}^J = q_t k_{f,t+1}^J + \text{PV of future markups}.$$

After a positive IST shock:

- assets in place still lose value because q_t falls;
- but the cost of future investment falls too;
- if firms have market power, lower future marginal cost raises the present value of future rents.

If the markup channel dominates the asset-in-place channel, then

$$\beta_M^z > 0.$$

Interpretation.

- Market power mainly affects the *exposure* of returns to IST shocks, not the SDF.
- This is why the paper says market power is mainly about the *beta*, not the *price of risk*.

3.5 EIS and the timing of uncertainty

The paper separates two preference margins:

- willingness to substitute consumption across time: $\text{EIS} = 1/\rho$;
- willingness to bear risk across states: γ .

Under CRRA utility:

- **high EIS** means households are willing to postpone consumption after good IST news, so current consumption falls more easily and the IST price of risk tends to be negative;

- **low EIS** means households strongly smooth consumption, so current consumption is supported and the IST price of risk tends to be positive.

Under recursive preferences:

- $\rho < \gamma$ means preference for **early resolution of uncertainty**;
- $\rho > \gamma$ means preference for **late resolution of uncertainty**.

A positive IST shock improves future utility. Therefore:

- with preference for early resolution, good IST news lowers the SDF and pushes λ^z up;
- with preference for late resolution, the opposite force appears.

Key subtle point.

- Flexible utilization and preference for early resolution can look similar in reduced form because both make good IST news reduce marginal utility.
- But they are conceptually different: utilization is a *technology* margin; early resolution is a *preference* margin.

3.6 The heatmap logic: when is the IST premium positive?

Figure 5 puts everything together by plotting the IST price of risk, the IST beta, and the IST contribution to the risk premium as functions of utilization flexibility and market power.

- With **late-resolution / low-EIS** preferences, positive IST premia arise either from the traditional channel (fixed utilization, negative λ^z , negative β_M^z) or from the new channel (flexible utilization plus high market power, positive λ^z , positive β_M^z).
- With **early-resolution / high-EIS** preferences, the price of risk is already positive in much of the state space; then the key remaining condition is enough market power to turn β_M^z positive.

This is the paper’s conceptual contribution. A positive risk premium on IST shocks does *not* uniquely imply a negative price of risk and negative beta. The same premium can come from the opposite sign pattern once one allows for utilization and rents.

4 Calibration and quantitative discipline (what pins down the numbers)

4.1 Monthly calibration and empirical targets

The model is calibrated at the **monthly** frequency and then evaluated on **annual** moments.

- Macro and asset-pricing moments are matched using U.S. data, mainly from **1930–2012**.
- Labor starts in **1947**; capacity utilization for the U.S. industrial sector starts in **1967**.
- To compare model returns to equity data, the authors scale model returns by a leverage factor of **1.5**.

4.2 Benchmark parameter values (Table 1)

The benchmark calibration uses:

- **Preferences:** annual $\beta^{12} = 0.98$, $\gamma = 10$, $EIS = 2$, $\theta = 1.6$, and ψ chosen so steady-state hours are **23%** of available time.
- **Production:** annual depreciation $12\delta_0 = 10\%$; labor shares $\alpha^C = \alpha^I = 0.60$; adjustment-cost curvatures $\phi^C = 14$ and $\phi^I = 7$; utilization-curvature parameters $\xi^C = 1.1$ and $\xi^I = 0.6$.
- **Market power:** $\nu^C = \nu^I = 4$, implying a gross markup of $4/3$, i.e. a net markup of about **33%**.
- **Technology shocks:** annual TFP growth 0.10% with volatility 1.60% ; annual IST growth 3.00% with volatility 2.60% .
- **Long-run risks:** annual persistence $\rho_{\mu^a}^{12} = \rho_{\mu^z}^{12} = 0.77$, and long-run volatility equal to **10%** of short-run volatility for both shocks.

Interpretation.

- The benchmark bakes in exactly the environment the theory says should matter most for a sizeable premium: *market power*, *some utilization flexibility*, and *high EIS / early-resolution preferences*.
- The asymmetric adjustment-cost parameters, $\phi^C > \phi^I$, help the model match the much greater volatility of investment growth than consumption growth.

4.3 Long-run IST risk: an important empirical input

Before calibrating, the authors document that IST shocks contain a small but persistent long-run component, in the same spirit as the long-run component in TFP emphasized by Croce.

- They use the relative price of capital goods as the main empirical IST series.
- Using the Bansal–Yaron style decomposition, they estimate annual persistence of the long-run IST component around **0.86**.
- The long-run IST volatility is about **26%** of short-run IST volatility in the data.

Why this matters.

- The benchmark calibration does not use the full empirical persistence/volatility estimates; instead it chooses a conservative common persistence of 0.77 and relative long-run volatility of 10% .
- That makes the quantitative success of the model more informative: the model does not need extreme long-run-risk calibration to work.

4.4 Numerical solution and accuracy

The rescaled model is solved on a four-dimensional state space:

- μ_t^a ,

- μ_t^z ,
- relative sectoral capital k_t ,
- a rescaled I-sector productivity-capital ratio state.

The authors use value-function iteration (VFI), not a local perturbation method.

- Grid: 30 points each for the two continuous endogenous states, 5 points each for μ_t^a and μ_t^z , and 5-point Gauss–Hermite quadrature for each i.i.d. shock.
- Maximum Euler equation error: about **1.97%** on the state space and **0.59%** in simulations.
- Average absolute Euler error in simulations: below **0.09%**.

Important implementation point.

- A second-order perturbation solution gives wildly different asset-pricing moments: mean annual log r_f around **-2.38%** and mean annual log premium around **14.36%**, versus **0.50%** and **5.14%** under VFI.
- So the paper’s quantitative claims rely heavily on using a global solution method.

4.5 What the benchmark matches well

Table 2 shows that the benchmark model matches many macro moments reasonably well:

- mean consumption growth: data **2.01%**, model **2.19%**;
- mean investment growth: data **2.22%**, model **2.14%**;
- mean output growth: data **1.93%**, model **2.13%**;
- mean growth of the relative price of investment: data **-3.45%**, model **-3.08%**;
- mean annual log risk-free rate: data **0.54%**, model **0.50%**;
- mean annual log equity premium: data **5.48%**, model **5.14%**.

It also does a decent job on the volatility of investment growth and capacity utilization.

4.6 What the benchmark misses

The paper is explicit about its limitations.

- **Return predictability.** In the data, the price-consumption ratio predicts *returns* but not *consumption growth*. In the model, it predicts consumption growth strongly and returns weakly.
- **Dividend cyclical.** Model dividends are too countercyclical. Simulated dividend growth has mean about **1.87%**, volatility about **16.22%**, and correlation with output growth about **-0.48**.
- **Equity volatility.** The model matches the mean premium much better than the volatility of equity returns; model excess-return volatility is only **6.08%** versus about **19.87%** in the data.

Interpretation.

- The paper is stronger on the *level and sign* of risk premia than on time-series forecasting moments.
- That is common in production-based long-run-risk models, but it is still an important caveat.

5 Calibration credibility and limitations (what makes the quantitative results persuasive)

- **The model isolates the mechanisms cleanly.** Capital utilization, market power, and preferences enter in distinct places. That makes the comparative statics unusually transparent.
- **The benchmark is disciplined by both macro and finance moments.** The paper does not just fit the equity premium; it also tries to match consumption, investment, labor, utilization, and the relative price of capital goods.
- **The long-run-risk assumption is empirically motivated.** The persistent components in TFP and IST are not inserted out of nowhere; the paper presents evidence that both exist in the data.
- **The decomposition of risk premium is informative.** By splitting the premium into prices of risk and betas, and then into short-run versus long-run shocks, the paper makes it possible to see *why* a given calibration works.
- **The global solution matters.** The authors show that local perturbation delivers distorted asset-pricing moments, so the choice of VFI increases credibility.

Main limitations.

- The model still struggles with return predictability and dividend dynamics.
- Market power is imposed in reduced form through CES differentiation and constant markup rather than endogenized from richer industrial-organization microfoundations.
- The quantitative role of utilization is sensitive to preferences and the persistence structure of shocks; the sign intuition from the simple CRRA case does not carry over mechanically to the benchmark with recursive preferences and long-run risk.
- The leverage factor of 1.5 is a practical bridge from claims on capital to observed equity returns, but it is not derived inside the model.

6 Tables and figures

6.1 Figure 1: flexible capital utilization

Read: This figure compares IRFs to a one-standard-deviation IST shock under *flexible* versus *fixed* capital utilization, holding preferences at log utility and setting market power essentially to zero.

- With fixed utilization, consumption falls, the SDF rises, and market value falls.
- With flexible utilization, consumption rises (or falls much less), the SDF falls, and market value still falls under competition but by less.

Why it matters. This is the clean visual proof that utilization mainly works through the *price of risk*. The sign of λ^z is not robust to whether old capital can be used harder.

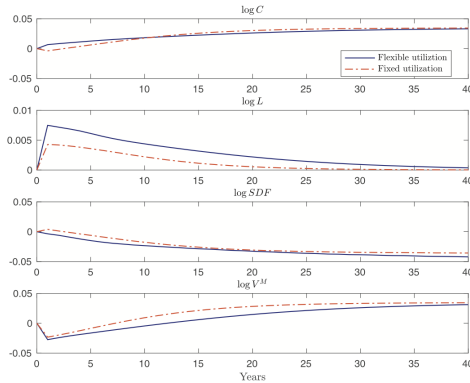


Fig. 1. The effect of flexible capital utilization. The figure plots the impulse response functions (IRFs) of consumption ($\log C$), SDF ($\log SDF$), and market portfolio value ($\log V^M$) to one-standard-deviation shock to the investment-specific technology. The utility is a log function ($\rho = \gamma = 1$) and firms do not have market power ($v_1 = v_2 = 10,000$). For each variable, the figure reports the IRFs under two parameterizations: (i) capital utilization is flexible ($\xi^c = \xi^l = 0.3$), solid line; (ii) capital utilization is fixed ($\xi^c = \xi^l = 3,000$), dashed line. The capital adjustment cost parameter $\phi^c = \phi^l = 2$. All the other parameter values are the same as the annualized values in Table 1.

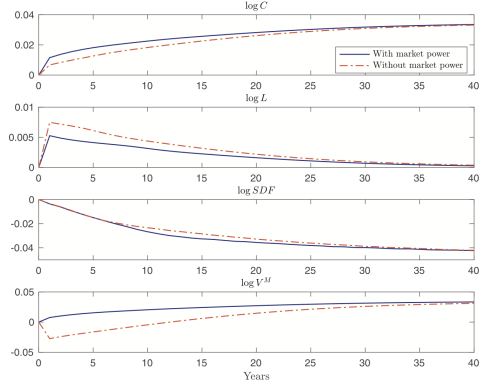


Fig. 2. The effect of market power. The figure plots the impulse response functions (IRFs) of consumption ($\log C$), SDF ($\log SDF$), and market portfolio value ($\log V^M$) to one-standard-deviation shock to the investment-specific technology. The utility is a log function ($\rho = \gamma = 1$) and capital utilization is flexible ($\xi^c = \xi^l = 0.3$). For each variable, the figure reports the IRFs under two parameterizations: (i) firms have market power ($v_1 = v_2 = 4$), solid line; (ii) firms are perfectly competitive ($v_1 = v_2 = 10,000$), dashed line. The capital adjustment cost parameter $\phi^c = \phi^l = 2$. All the other parameter values are the same as the annualized values in Table 1.

6.2 Figure 2: market power

Read: This figure holds utilization flexible and compares perfect competition to monopolistic competition.

- Consumption and the SDF move similarly across the two cases.
- The big difference is the response of market value.
- Under competition, firm value falls after positive IST news.
- Under sufficient market power, firm value rises.

Why it matters. This is the paper's cleanest evidence that market power mainly works through the *beta*, not through the SDF.

6.3 Figures 3 and 4: EIS and temporal resolution

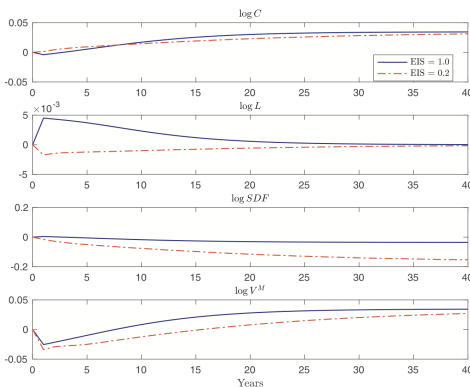


Fig. 3. The effect of EIS. The figure plots the impulse response functions (IRFs) of consumption ($\log C$), SDF ($\log SDF$), and market portfolio value ($\log V^M$) to one-standard-deviation shock to the investment-specific technology. Capital utilization is fixed ($\xi^c = \xi^l = 3,000$), and firms are perfectly competitive ($v_1 = v_2 = 10,000$). For each variable, the figure reports the IRFs under two parameterizations: (i) high EIS ($\xi^c = 1$), solid line; (ii) low EIS ($\xi^c = 0.2$), dashed line. The capital adjustment cost parameter $\phi^c = \phi^l = 2$. All the other parameter values are the same as the annualized values in Table 1.

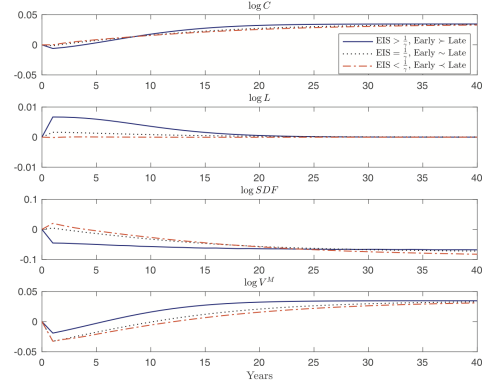


Fig. 4. The effect of preference towards temporal resolution of uncertainty. The figure plots the impulse response functions (IRFs) of consumption ($\log C$), SDF ($\log SDF$), and market portfolio value ($\log V^M$) to one-standard-deviation shock to the investment-specific technology. Capital utilization is fixed ($\xi^c = \xi^l = 3,000$), and firms are perfectly competitive ($v_1 = v_2 = 10,000$). For each variable, the figure reports the IRFs under three parameterizations: (i) Preference towards early resolution of uncertainty ($\rho < \gamma$, with $\rho = 0.5$ and $\gamma = 2$), solid line; (ii) Indifference between early vs. late resolution of uncertainty ($\rho = \gamma = 2$), dotted line; and (iii) Preference towards late resolution of uncertainty ($\rho > \gamma$, with $\rho = 3$ and $\gamma = 2$), dash-dotted line. The capital adjustment cost parameter $\phi^c = \phi^l = 2$. All the other parameter values are the same as the annualized values in Table 1.

Read: Figure 3 changes the EIS under CRRA utility; Figure 4 changes preference for early versus late resolution under recursive utility.

- In Figure 3, low EIS pushes households to smooth consumption more strongly, which can make the IST price of risk positive even without flexible utilization.
- In Figure 4, preference for early resolution makes good IST news lower marginal utility, which also supports a positive price of risk.

Why it matters. These figures show that technology and preferences can be observationally similar in asset-pricing space. One has to separate them carefully.

6.4 Figure 5: heatmap summary of the whole theory

Read: This is the master figure. It plots the IST price of risk, the IST market beta, and the IST risk premium over a grid of utilization flexibility and market power, separately for low-EIS and high-EIS economies.

- In low-EIS / late-resolution economies, positive IST premia can arise in two qualitatively different regions.
- In high-EIS / early-resolution economies, the price of risk is often already positive, so the main issue is whether market power is high enough to make the beta positive too.

Why it matters. This figure explains how the paper reconciles competing views in the literature. The same positive premium can come from opposite sign combinations of λ^z and β_M^z .

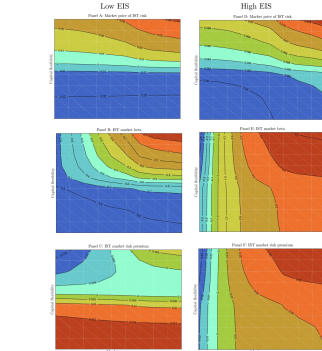


Fig. 6. Asset pricing implications of capital flexibility, market power, and EIS. The figure plots the heatmap of these variables as a function of capital flexibility, market power, and market power. (1) IST market price of risk, (2) IST market beta, and (3) IST risk premium. The heatmaps are plotted for the parameter values $\lambda^z = 1$, $\beta_M^z = 1$, and $\beta^z = 1$. The heatmaps are plotted for the parameter values $\lambda^z = 1$, $\beta_M^z = 1$, and $\beta^z = 1$. The heatmaps are plotted for the parameter values $\lambda^z = 1$, $\beta_M^z = 1$, and $\beta^z = 1$.

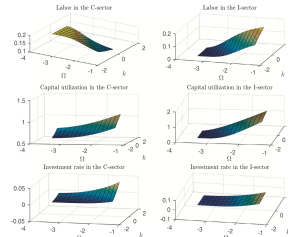


Fig. 7. Equilibrium free policies. The figure plots the equilibrium labor, investment rate, and capital utilization in the C- and I-sectors, as a function of the state variables k and z defined in (3.4). The expected growth rate μ^k and μ^z are set at their respective unconditional averages. Parameter values are given in Table 1.

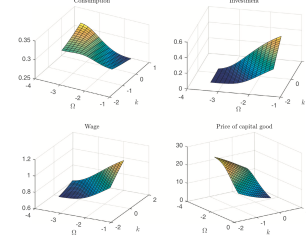


Fig. 8. Equilibrium aggregate policies. The figure plots the equilibrium consumption, investment, wage, and price of capital good, as a function of the state variables k and z defined in (3.4). The expected growth rate μ^k and μ^z are set at their respective unconditional averages. Parameter values are given in Table 1.

6.5 Figures 6 and 7: equilibrium policy functions

Read: These figures show how sectoral labor, investment, utilization, consumption, investment, wages, and the capital-good price vary over the state space.

- When the I-sector is relatively productive or the C-sector capital stock is relatively low, I-sector labor rises.
- Investment and utilization rise in both sectors in those states.

- Wages rise while the price of capital falls.

Why it matters. These figures connect the intuitive IRF story to the full nonlinear equilibrium. They show where the pricing results come from in the policy functions, not just in local impulse responses.

6.6 Table 1: benchmark calibration

Read: Table 1 lists the preference, production, and shock parameters.

- High EIS (= 2) and high risk aversion ($\gamma = 10$) create strong early-resolution effects.
- $\nu = 4$ implies about a 33% markup.
- $\xi^C = 1.1$ and $\xi^I = 0.6$ imply that utilization is flexible, but not as flexible as in some business-cycle calibrations.

Why it matters. These parameter choices are exactly where the paper’s mechanisms are supposed to matter most.

Table 1
Parameter values used in the benchmark monthly calibration of Section 4.

Group	Description	Symbol	Value
Preferences	Time discount rate	β^{12}	0.98
	Relative risk aversion	γ	10.0
	EIS	$1/\rho$	2.0
	Degree of labor disutility	θ	1.6
	Sensitivity of labor disutility	ψ	2.957 ^a
Production	Depreciation rate constant	$12 \times \delta_0$	0.10
	Depreciation rate slope	δ_1	0.0134 ^b
	Labor share of output for C-sector	α^C	0.60
	Labor share of output for I-sector	α^I	0.60
	Degree of capital adjustment cost for C-sector	ϕ^C	14.00
	Degree of capital adjustment cost for I-sector	ϕ^I	7.00
	Elasticity of marginal depreciation for C-sector	ξ^C	1.10
	Elasticity of marginal depreciation for I-sector	ξ^I	0.60
	Constant elasticity of substitution for C-sector	ν_C	4.00
	Constant elasticity of substitution for I-sector	ν_I	4.00
	Financial leverage	...	1.50
Shocks	Growth rate of TFP shock	$12 \times \bar{\mu}^a$	0.10%
	Volatility of TFP shock	$\sqrt{12} \times \sigma_a$	1.60%
	Volatility of long-run TFP risk	$\sqrt{12} \times \sigma_{\mu^a}$	0.16%
	Persistence of long-run TFP risk	$\rho_{\mu^a}^{12}$	0.77
	Growth rate of IST shock	$12 \times \bar{\mu}^z$	3.00%
	Volatility of IST shock	$\sqrt{12} \times \sigma_z$	2.60%
	Volatility of long-run IST risk	$\sqrt{12} \times \sigma_{\mu^z}$	0.26%
	Persistence of long-run IST risk	$\rho_{\mu^z}^{12}$	0.77

^a Chosen such that the fraction of working hours is 23% of available time in the deterministic steady-state.

^b Chosen such that capital utilization equals to one in the deterministic steady-state.

Table 2

Model versus data: macroeconomic and asset pricing moments.

This table compares macroeconomic and asset pricing moments of the data to simulated moments from the model (in percentage). The empirical moments are estimated from the U.S. annual sample from 1930–2012 (the labor time series starts in 1947, and capacity utilization starts in 1967). The log growth rates of consumption (Δc), investment (Δi), total output (Δy) are adjusted for inflation and population. The log growth rate of labor (Δl) is adjusted for population. The growth rate of relative price of investment good (Δp^I) is adjusted for quality. The log growth rate in capital utilization (Δu) in the model is based on the average of the two sectors. r_f is the log risk-free rate adjusted for inflation and r_M^{ex} is the log market excess return, i.e., the log market return in excess of the log risk-free rate. For the empirical moments, we report both the point estimates and the 95%-confidence intervals. For the model-implied moments, we report the median of 1,000 simulations as the point estimates and the 2.5- and 97.5-percentiles for the 1,000 simulations. Each simulation consists of a time series of 100 years, constructed from a monthly series. Further details on the data constructions are provided in Appendix C.

Variable	Data		Model	
	Mean	Volatility	Mean	Volatility
<i>Panel A: Macroeconomic quantities</i>				
Δc	2.01	2.22	2.19	3.91
	[1.53, 2.50]	[1.92, 2.62]	[−0.67, 4.96]	[2.96, 5.11]
Δi	2.19	13.58	2.14	13.67
	[−0.77, 5.15]	[11.78, 16.02]	[−1.30, 5.35]	[10.47, 20.57]
Δl	−0.02	2.62	−0.00	2.10
	[−0.67, 0.63]	[2.23, 3.17]	[−0.21, 0.20]	[1.70, 2.45]
Δy	1.93	5.58	2.13	5.11
	[0.71, 3.15]	[4.84, 6.58]	[−0.68, 4.99]	[4.03, 6.34]
Δp^I	−3.45	3.65	−3.08	4.02
	[−4.25, −2.65]	[3.17, 4.31]	[−5.27, −0.67]	[3.16, 5.08]
Δu	−0.26	4.21	0.00	3.12
	[−1.53, 1.00]	[3.49, 5.32]	[−0.47, 0.44]	[2.51, 3.81]
<i>Panel B: Asset prices</i>				
r_f	0.54	2.85	0.50	1.08
	[−0.09, 1.17]	[2.47, 3.36]	[−0.32, 1.33]	[0.79, 1.48]
r_M^{ex}	5.48	19.87	5.14	6.08
	[1.14, 9.82]	[17.24, 23.46]	[3.22, 7.35]	[4.98, 7.20]

6.7 Table 2: model versus data

Read: Table 2 compares simulated and empirical annual moments.

- The model gets the *means* of macro growth rates and the average equity premium roughly right.
- The largest miss is that the model’s market-return volatility (**6.08%**) is far below the data (**19.87%**).

Why it matters. The paper should be read as explaining the *sign and level* of the premium, not as a full quantitative match to all return moments.

6.8 Table 3: return predictability versus consumption-growth predictability

Read: This table compares long-horizon predictive regressions in the data and the model.

- In the U.S. data, the price-consumption ratio strongly predicts excess returns: the slope goes from about **-11.96** at the one-year horizon to about **-43.79** at the five-year horizon.
- In the model, return predictability slopes are weak and even positive, while consumption-growth predictability is extremely strong: for example, the one-year β_C is about **35.61**, rising to **135.01** at five years.

Why it matters. This is the sharpest quantitative failure of the model and an important caveat for how far one should push its time-series implications.

6.9 Table 4: decomposition of the aggregate market premium

Read: This table decomposes the market premium into prices of risk and betas for four shocks: short-run TFP, short-run IST, long-run TFP, and long-run IST.

Benchmark (Panel A):

- total premium $RPM = 5.366\%$;
- short-run TFP contribution: **0.559%** (**10.4%**);
- short-run IST contribution: **0.024%** (**0.4%**);
- long-run TFP contribution: **4.172%** (**77.8%**);
- long-run IST contribution: **0.611%** (**11.4%**).

Counterfactuals:

- **No market power, flexible utilization (Panel B):** $RPM = -0.624\%$.
- **Market power, fixed utilization (Panel C):** $RPM = 6.219\%$.
- **No market power, fixed utilization (Panel D):** $RPM = -0.257\%$.
- **Low EIS (Panel E):** $RPM = 0.559\%$.

Why it matters. This is the core quantitative table. It shows that long-run risks dominate, market power is essential, and low EIS collapses the premium.

Table 3

Predictability of returns and consumption growth rates.

This table reports the results of the long-horizon predictability of future excess returns and consumption growth rates. The long-horizon predictive regression of excess return is $\sum_{t=1}^H (R_{t+H,t} - R_{t,t}) = \alpha_H + \beta_H \log(R_t/G_t) + \epsilon_{t+H}^H$, where H is the forecasting horizon in years, $R_{t+H,t}$ is the real stock market return, and $R_{t,t}$ is the real risk-free rate. P_t is the real Standard & Poor's (S&P) 500 stock market index, and G_t is real consumption. The long-horizon predictive regression of log consumption growth is $\sum_{t=1}^H \log(G_{t+H,t}/G_t) = \alpha_H + \beta_H \log(R_t/G_t) + \epsilon_{t+H}^H$. Real consumption is defined as real per capita nondurables plus services from NIPA Table 7.1. The real S&P500 stock market index is obtained by deflating the nominal index from the Center for Research in Security Prices (CRSP). Return data are from Kenneth French's website. The entries for the model are the average estimates based on 1,000 simulations each with 1,200 monthly observations that are time-aggregated to an annual frequency. The slopes and R^2 s are in percent. We report in brackets the 95%-confidence interval for the slopes. In constructing the confidence intervals in the data, the standard errors are Newey and West (1997) corrected with $2(H-1)$ lags.

U.S. annual data, 1930–2015					
H	1y	2y	3y	4y	5y
β_x	-11.96 [-20.80, -3.12]	-19.18 [-32.96, -5.40]	-28.65 [-44.62, -12.67]	-37.63 [-55.02, -20.24]	-43.79 [-65.52, -22.06]
R^2	7.12	9.79	16.95	22.94	25.45
β_c	-0.57 [-1.43, 0.29]	-1.21 [-3.09, 0.66]	-1.77 [-4.61, 1.08]	-2.05 [-5.90, 1.80]	-2.02 [-6.82, 2.78]
R^2	1.36	2.52	3.84	4.03	3.34
Model					
H	1y	2y	3y	4y	5y
β_x	8.28 [-10.24, 20.43]	14.37 [-19.11, 38.14]	18.39 [-31.36, 54.81]	20.86 [-42.11, 71.08]	22.00 [-56.01, 94.78]
R^2	3.52	4.99	5.66	5.99	6.13
β_c	35.61 [30.89, 40.76]	66.86 [56.72, 75.66]	93.38 [76.12, 106.67]	115.95 [89.08, 135.55]	135.01 [97.84, 162.77]
R^2	67.66	75.06	75.66	73.94	68.99

Table 4

Decomposition of aggregate market risk premium.

This table reports the decomposition of market risk premia into prices of risk and betas of the two short-run shocks, ϵ_t^a , ϵ_t^z , and the two long-run shocks, $\epsilon_t^{\mu^a}$, and $\epsilon_t^{\mu^z}$. Panel A represents the benchmark calibration in Table 2 with flexible capital utilization and market power. In Panel B, capital utilization is flexible and firms are perfectly competitive. In Panel C, capital utilization is fixed and firms have market power. In Panel D, capital utilization is fixed and firms are perfectly competitive. In Panel E, EIS has a low value of 0.75. All the other parameters are kept to their benchmark levels contained in Table 1. For each combination of alternative parameters, we simulate the model in the same way as in the benchmark calibration of Table 2, and use the full simulation population to estimate the following variables: (i) price of risk, λ^x , (ii) market return loading, β_M^x , and (iii) leveraged risk premium, $1.5 \times \beta_M^x \lambda^x$, for each shock and the summation of all shocks, which is the market risk premium, RP_M . Both the price of risk and risk premium are in percentage terms.

	Shock (X)				Total RP_M
	Short-run		Long-run		
	ε_t^a	ε_t^z	$\varepsilon_t^{\mu^a}$	$\varepsilon_t^{\mu^z}$	
Panel A: Flexible capital utilization and market power					
λ^x	0.349	0.245	0.140	0.090	
β_M^x	1.070	0.065	19.816	4.524	
$1.5 \times \beta_M^x \lambda^x$	0.559	0.024	4.172	0.611	5.366
Contribution	10.4%	0.4%	77.8%	11.4%	100%
Panel B: Flexible capital utilization and perfect competition					
λ^x	0.334	0.207	0.135	0.077	
β_M^x	0.492	-0.510	3.906	-12.962	
$1.5 \times \beta_M^x \lambda^x$	0.246	-0.158	0.788	-1.501	-0.624
Contribution	-39.5%	25.3%	-126.3%	240.5%	100%
Panel C: Fixed capital utilization and market power					
λ^x	0.362	0.280	0.147	0.106	
β_M^x	1.102	0.097	22.003	4.658	
$1.5 \times \beta_M^x \lambda^x$	0.598	0.041	4.837	0.743	6.219
Contribution	9.6%	0.7%	77.8%	12.0%	100%
Panel D: Fixed capital utilization and perfect competition					
λ^x	0.351	0.252	0.142	0.096	
β_M^x	0.571	-0.431	6.201	-11.887	
$1.5 \times \beta_M^x \lambda^x$	0.301	-0.163	1.319	-1.714	-0.257
Contribution	-117.0%	63.5%	-513.4%	667.0%	100%
Panel E: Low EIS					
λ^x	0.338	0.218	0.136	0.095	
β_M^x	0.754	-0.250	-2.169	4.906	
$1.5 \times \beta_M^x \lambda^x$	0.382	-0.082	-0.442	0.701	0.559
Contribution	68.4%	-14.6%	-79.2%	125.4%	100%

6.10 Table 5: varying capital-utilization flexibility

Read: This table changes $\xi^C = \xi^I = \xi$, so moving down the table makes utilization less flexible and approaches the fixed-utilization case.

- In Panel A, the total market premium *increases* as utilization becomes *less* flexible: from **5.550%** at $\xi = 1.0$ to **6.199%** at $\xi = 50$.
- That seems to contradict the simple CRRA intuition from Section 3.

Why the sign flips.

- Panel B isolates the contribution from *period utility*; those prices of risk fall as utilization becomes less flexible, just as the simple intuition says.
- Panel C isolates the contribution from *continuation utility*; those prices of risk rise as utilization becomes less flexible.
- In the benchmark with recursive preferences and long-run risk, the continuation-utility effect dominates.

Why it matters. This is the subtlest table in the paper. It shows that one cannot mechanically extrapolate from the short-run CRRA logic to the benchmark quantitative model.

Table 5
Asset pricing implications of capital utilization flexibility.

This table reports the risk premium, RP_M , and its decomposition under different degrees of flexibility in capital utilization. We set $\xi^C = \xi^I = \xi$. Panel A reports results when all the other parameters are set to the benchmark values in Table 1. In Panels B and C we apply the decomposition of SDF in Eq. (31) to the prices of risk reported in Panel A. Specifically, Panel B reports the component of price of risk that derives from period utility. Panel C reports the component of price of risk that derives from continuation utility. The risk exposures (β^i) in Panels B and C are the same as those in Panel A. Panel D reports results when firms are perfectly competitive. Both the price of risk and risk premium are in percentage terms.

ξ	RP_M	Price of risk				Exposure			
		λ^A	λ^I	λ^{P^*}	λ^{P^*}	β_M^A	β_M^I	$\beta_M^{P^*}$	$\beta_M^{P^*}$
Panel A: Market power and recursive preferences									
Flexible: 1.0	5.550	0.351	0.251	0.141	0.093	1.091	0.087	20.067	4.933
2.0	5.846	0.356	0.265	0.144	0.099	1.100	0.096	20.869	4.838
4.0	6.021	0.359	0.272	0.145	0.103	1.103	0.098	21.383	4.768
10.0	6.134	0.361	0.276	0.146	0.105	1.103	0.098	21.747	4.719
Fixed: 50.0	6.199	0.361	0.278	0.146	0.106	1.103	0.097	21.958	4.683
Panel B: Market power and recursive preferences: Risk premia from period utility									
Flexible: 1.0	0.031	0.014	0.003	-0.000	0.002	1.091	0.087	20.067	4.933
2.0	0.027	0.014	0.002	-0.000	0.001	1.100	0.096	20.869	4.838
4.0	0.024	0.013	0.001	-0.000	0.001	1.103	0.098	21.383	4.768
10.0	0.022	0.013	0.000	-0.000	0.000	1.103	0.098	21.747	4.719
Fixed: 50.0	0.021	0.013	-0.000	-0.000	0.000	1.103	0.097	21.958	4.683
Panel C: Market power and recursive preferences: Risk premia from continuation utility									
Flexible: 1.0	5.518	0.337	0.248	0.142	0.091	1.091	0.087	20.067	4.933
2.0	5.819	0.343	0.263	0.144	0.096	1.100	0.096	20.869	4.838
4.0	5.997	0.346	0.271	0.145	0.102	1.103	0.098	21.383	4.768
10.0	6.112	0.348	0.276	0.146	0.104	1.103	0.098	21.747	4.719
Fixed: 50.0	6.177	0.349	0.279	0.146	0.106	1.103	0.097	21.958	4.683
Panel D: Perfect competition and recursive preferences									
Flexible: 1.0	-0.630	0.338	0.216	0.136	0.081	0.916	-0.485	4.051	-12.720
2.0	-0.578	0.344	0.233	0.139	0.088	0.934	-0.467	4.726	-12.687
4.0	-0.472	0.348	0.243	0.140	0.092	0.949	-0.452	5.316	-12.420
10.0	-0.343	0.350	0.248	0.141	0.094	0.962	-0.440	5.823	-12.087
Fixed: 50.0	-0.273	0.351	0.251	0.142	0.096	0.969	-0.433	6.117	-11.924

Table 6
Asset pricing implications of market power and EIS.

This table reports the risk premium, RP_M , and its decomposition under different levels of market power and EIS. Panels A and B report the comparative statics for the effect of market power under high (2.0) and low (0.75) EIS, respectively. We set $\nu^C = \nu^I = \nu$. Panel C reports the comparative statics for the effect of EIS. All other parameters are set to the benchmark values in Table 1. Both the price of risk and risk premium are in percentage terms.

	RP_M	Price of risk				Exposure			
		λ^A	λ^I	λ^{P^*}	λ^{P^*}	β_M^A	β_M^I	$\beta_M^{P^*}$	$\beta_M^{P^*}$
Panel A: Effect of market power: EIS = 2.0									
ν									
Monopolistic: 3	6.084	0.352	0.254	0.142	0.093	1.130	0.125	21.426	6.310
4	5.366	0.349	0.245	0.140	0.090	1.070	0.065	19.816	4.524
11	2.802	0.339	0.220	0.137	0.082	0.936	-0.368	13.414	-2.580
21	1.538	0.337	0.214	0.136	0.080	0.712	-0.291	9.986	-6.362
Competitive: 10^4	-0.624	0.334	0.207	0.135	0.077	0.492	-0.510	3.906	-12.962
Panel B: Effect of market power: EIS = 0.75									
ν									
Monopolistic: 3	0.657	0.341	0.226	0.136	0.096	0.824	-0.179	-2.338	5.382
4	0.559	0.338	0.218	0.136	0.095	0.754	-0.250	-2.169	4.906
11	-0.634	0.331	0.201	0.135	0.091	0.547	-0.456	-4.399	0.883
21	-1.437	0.330	0.196	0.134	0.090	0.442	-0.561	-6.185	-1.803
Competitive: 10^4	-2.993	0.328	0.191	0.134	0.089	0.254	-0.747	-9.785	-7.008
Panel C: Effect of EIS									
EIS									
Low: 0.75	0.559	0.338	0.218	0.136	0.095	0.754	-0.250	-2.169	4.906
1.0	1.953	0.341	0.226	0.136	0.093	0.847	-0.158	4.139	5.185
1.5	3.944	0.346	0.237	0.138	0.091	0.962	-0.023	13.324	4.965
2.0	5.366	0.349	0.245	0.140	0.090	1.070	0.065	19.816	4.524
High: 2.25	5.898	0.349	0.245	0.141	0.089	1.101	0.096	22.332	4.209

6.11 Table 6: varying market power and EIS

Read: Table 6 is the clean comparative-static summary.

Panel A: varying market power with EIS = 2.0.

- Competitive markets: $RPM = -0.624\%$.
- Moderate market power, $\nu = 11$: $RPM = 2.802\%$.
- Benchmark, $\nu = 4$: $RPM = 5.366\%$.
- Strong market power, $\nu = 3$: $RPM = 6.084\%$.

Panel B: varying market power with low EIS = 0.75.

- Even with strong market power, the premium is small: **0.657%** at $\nu = 3$.
- As competition strengthens, the premium turns sharply negative, reaching **-2.993%** in the competitive limit.

Panel C: varying EIS.

- $RPM = 0.559\%$ at $EIS = 0.75$,
- $RPM = 1.953\%$ at $EIS = 1.0$,
- $RPM = 3.944\%$ at $EIS = 1.5$,
- $RPM = 5.366\%$ at $EIS = 2.0$,
- $RPM = 5.898\%$ at $EIS = 2.25$.

Why it matters. This table makes the ranking crystal clear: *market power* and *high EIS* are the strongest quantitative forces behind the premium.

7 Short summary

- **Method.** A two-sector production-based asset-pricing model with recursive preferences, variable capital utilization, monopolistic competition, and long-run risks in TFP and IST growth.
- **Core conceptual result.** Capital utilization mainly changes the *price of IST risk*; market power mainly changes the *market beta on IST shocks*.
- **Main quantitative result.** With market power, long-run risks, and high EIS, the model generates about a **5.14%** annual log equity premium; without market power, the premium turns negative.
- **Decomposition result.** Roughly **89%** of the premium comes from long-run technology risks.
- **Big lesson.** The sign of IST pricing objects is not structural by itself. It depends on how firms use old capital, how much rent they keep, and how households value intertemporal risk.

8 Conclusion

Core conclusion. Garlappi and Song show that the asset-pricing implications of investment shocks are fundamentally altered by two realistic features of the macroeconomy: firms can adjust the *utilization* of existing capital, and firms may possess *market power*. Once these features are introduced, the canonical negative IST price of risk and negative IST beta are no longer inevitable.

Economic intuition. Utilization flexibility changes how good investment news maps into *current consumption* and therefore into the SDF. Market power changes how good investment news maps into *firm value* by adding a monopoly-rent channel on top of the assets-in-place channel. Recursive preferences determine which of these margins dominate quantitatively.

Takeaway. The paper’s deepest contribution is not just that it matches the premium better. It shows exactly *which economic margins govern the sign of technology pricing*: utilization for the λ -side, market power for the β -side, and long-run recursive risk for the magnitude.